

Logitech MX Ink: Extending the Meta Quest Platform with a 6DoF Stylus

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ABSTRACT

There is a long history of research and development in the area of controller interactions for AR/VR platforms. However, in recent years the Meta Quest Touch Controller has emerged as a de facto reference design, and today most of the commonly available commercial 6DoF controllers are aligned with that core feature set. There are notable benefits for users, familiarity with one controller means it is easier to use a different (but similar) controller on another platform. There are also advantages for developers, as it (somewhat) standardizes the interaction mappings across different vendor platforms. The addition of a new type of input device to an existing ecosystem creates new opportunities, but also introduces challenges to existing paradigms which have been primarily focused on the platform standard controller. This workshop paper outlines some of the key challenges faced by app developers with the addition of the Logitech MX Ink Mixed Reality stylus to the Meta Quest platform.

Index Terms: Human-centered computing [Human-computer interaction (HCI)]; Interaction Devices—

1 INTRODUCTION

There is a decades-long history of research and development in interaction design for XR platforms [18]. Even though most of these explorations have remained as lab prototypes, today's mainstream XR platforms still offer a much more diverse set of input devices and interactions than are typically available on PC, laptop, tablet or smartphones.

In recent years the default platform interaction schemes have converged around the use of 6DoF tracked controllers, tracked hands/fingers, gaze and voice. This core set of interaction methods can be used to navigate and interact within the XR operating system home environment and apps. The SDKs, frameworks and documentation provided as part of the XR platform devkit supports these default methods, thus making it easier to develop apps.

A stylus is a long-established device input category on PCs, tablets and smartphones. Over time, the basic RAND tablet concept [7] was enhanced with additional capabilities. Today, even in a 2D tablet scenario, the Apple Pencil APIs can report tilt (altitude), force (pressure), orientation (azimuth) and barrel roll. Some styli also support hover interactions above the touch surface. But even with the addition of more capabilities and additional DoFs, these interactions are very closely coupled with use on a planar 2D touch surface.

A 6DoF stylus is also a category of input device with a long research history. Attempts to use passively tracked stylus approaches did not yield acceptable performance; solutions based on higher end tracking technologies such as OptiTrak were not a mainstream option for cost reasons [5]. The HTC Vive product, using the SteamVR

Lighthouse tracking system, facilitated new explorations with improved tracking precision and accuracy on consumer XR systems [3]. The SteamVR tracking system enabled stylus prototypes that could be (somewhat) closer to a traditional stylus form factor that users are already familiar with from 2D tablets [10]. The Logitech VR Ink Pilot Edition was one of such devices [15]. This device was used in several academic studies exploring controller grips [4] and 6DoF sketching performance [6].

The Logitech MX Ink product for Meta Quest was released in September 2024 [14]. It extends the Meta Quest platform with the addition of a 6DoF stylus. This workshop paper describes some of the challenges in introducing this 6DoF stylus device to the Meta Quest ecosystem.

2 ADDING BASIC STYLUS SUPPORT

The MX Ink stylus is natively supported in the Meta Quest platform, no additional hardware or software components must be installed. The stylus is paired with the HMD in a manner similar to the default platform Touch controllers. There is an MX Ink OpenXR interaction profile defined, which allows app developers to detect the stylus, and thus support stylus-specific functionality. There is also a fallback mode, in which the stylus operates as a standard Meta Touch controller with a subset of controls, thus allowing for a certain amount of backward compatibility in apps that do not specifically support the stylus interaction profile. The input and output of the stylus are described in Fig. 1.

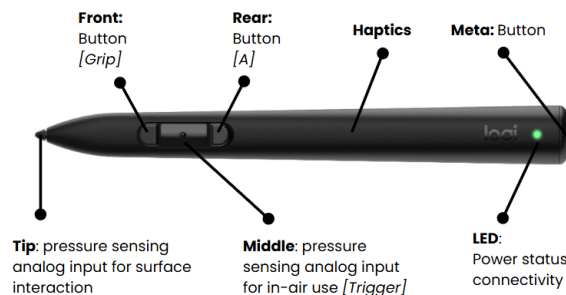


Figure 1: MX Ink Overview

Beyond pairing, tracking and supporting the OpenXR interaction profile, there are numerous other considerations for developers as they support this new device category. This section lists some of the design and implementation issues when adding stylus support to an app.

2.1 Developer APIs

It is important to provide SDKs and references to allow developers to add support for a new category of input devices. Developers for Meta Quest use a wide range of development tools and languages including Unity, Unreal and WebXR. The MX Ink GitHub page includes details on how to add stylus support in an app for these different environments [14].

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2.2 Coexistence with Controllers and Hands

Today the Meta Quest ships with two 6DoF Touch controllers; every user that purchases the Meta Quest will have these. The Quest platform also supports hand tracking, and the tracked hands can be used for both direct and indirect interaction using raycast, poke, pinch and some custom gestures. The stylus coexists with both the Touch and hand controls. The user can dynamically switch between controllers, stylus and hand interactions as needed, even within an app. Sample code to support this dynamic device detection and swapping of input devices is included in the GitHub repository.

2.3 Different Axis and Button Inputs

The stylus is not a direct replacement for the controller. The MX Ink stylus has a significantly different control set from the Meta Touch Controller, as shown in Figure 1. The Middle pressure-sensitive button is typically mapped to the Touch Controller Trigger input, the Front button is typically mapped to the Touch Controller Grip input, the Rear button is typically mapped to the Touch Controller "A" button. The stylus does not have an equivalent to the Touch Controller "B" button or Thumbstick. However, the stylus also has a pressure-sensitive tip that can be used for interaction with 2D surfaces. This different set of inputs means that for more complex apps, it is often more than just a simple remapping of inputs to support a stylus. Some of the approaches to address this are as follows:

- **Additional UI Elements.** The app may add (or already have) UI driven methods to perform the equivalent function, e.g., Open Brush brush stroke size can be adjusted using the Touch Controller Thumbstick, or via the UI brush slider as shown in Fig. 2 (a).
- **Additional Spatial UI/Gesture Controls.** Vermillion added a spatial interaction to change brush parameters using the stylus. This can also be done using the Touch Controller Thumbstick.
- **Move or Duplicate Controls to NDH Touch Controller.** Some applications may have unused inputs on the NDH Touch Controller, and actions can be reassigned to these unused inputs, as shown in Fig. 2 (b).

Beyond the mapping of inputs, there are additional differences in how the analog inputs for the Touch Controller Trigger and the MX Ink pressure-sensitive controls work. [16]. The Touch Controller trigger typically has a large deadzone before inputs are reported, i.e. the user must move the trigger a certain distance to generate inputs. In contrast, the MX Ink reports inputs with much lower force values. As a result, for certain use cases, such as pressure-sensitive control of line width, the processing of input for both devices can be different.

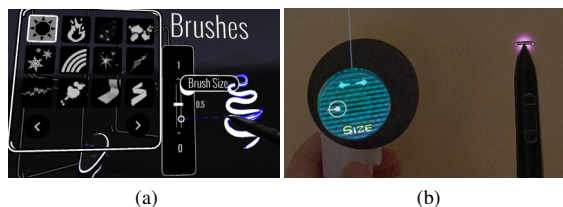


Figure 2: (a) Brush size slider added in Open Brush (b) FigminXR moved brush adjustment to non-dominant hand

2.4 Supporting Haptics

Haptic feedback is an integral part of the Meta Quest OS experience. For example, in the home environment, there is visual and haptic feedback associated with many UI interactions. Meta Haptics Studio and the Haptics SDK tools are the recommended path to create and integrate haptics in an app. The MX Ink stylus has an integrated linear resonant actuator (LRA) that is compatible with the Meta Quest haptics tools. To date, our experience is that the same haptic effects can work well across both the MX Ink stylus and Touch controllers.

3 STYLUS OPTIMIZATIONS

There are a range of additional design and development considerations to fully exploit the potential of a stylus.

3.1 Handedness

Developers must dynamically support stylus handedness. Most 6DoF controllers are designed for use with either the left-hand or the right-hand. The HTC Vive wands are a notable exception in being a symmetric design. These devices are dynamically assigned to the right and left when the XR system software initializes. The MX Ink device is also hand-agnostic, and it can be used in either hand. The OpenXR interaction profile definition is such that it will work with either hand from a software perspective. There are several cases in which the developer will want to know which hand is holding the stylus and adjust the UI accordingly. A typical example of this is the commonly supported feature of an extensible UI panel associated with the non-dominant hand; this will expand to the right or the left depending on the handedness of the user. Fig. 3 shows Gravity Sketch Settings UI Panel expanded for a right-hand user.



Figure 3: Expandable UI Panel in Gravity Sketch

3.2 Stylus Model Options

Apps typically show a visual representation of the controller being used. The most common approach is to render a visual representation of the real-world physical controller, that is, the rendered model overlays the real-world physical controller. Based on our work with early MX Ink developers, they have implemented a range of other approaches optimized for a stylus, including:

- **Transparent Stylus Model.** This can be a valuable feature in certain fully immersive use cases. The transparent stylus model does not completely obscure the content that is being

created. Fig. 4 (a) shows an example of this approach from the GestureVR app. In this case, it is possible to see through the stylus model and view the content on the virtual drawing board.

- **Stylus Tip Only.** Only showing the tip of the stylus or the interaction point maximizes the available space for content and minimizes the visual space taken up by the stylus model. Fig. 4 (b) is a screenshot from the Quill app. There is no controller model shown. The dot on the right-hand side of the image represents the stylus tip inking point.

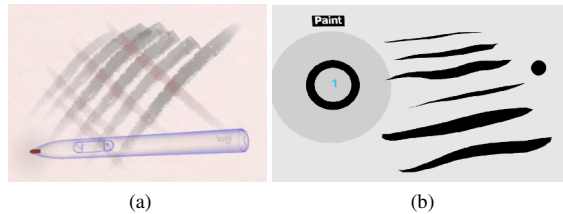


Figure 4: (a) Gesture VR transparent stylus model (b) Quill shows the inking point, the dot at the extreme right of the image, but no controller model

3.3 Stylus Grip Representation (or none!)

In addition to the visual representation of the stylus, it is also important to consider issues associated with the visual representation of the user hands. This is especially important in fully immersive use cases where the user cannot see their real hands. The accuracy of the virtual hand representation is important to users [1]. The grip for the Meta Touch controllers is very prescriptive. These constraints, together with the touch-sensing inputs on buttons, make it easier to render a virtual hand and finger representation that is very close to the real-world.

In contrast, a large number of different grips for a stylus are typically observed when using a stylus in 2D writing and sketching [20]. Many more additional grips can be observed when using a stylus in 3D space [13]. This large range of possibilities makes it very difficult to accurately render a model of the user's hand holding the stylus. Some examples are shown in Fig. 4, where (a) shows the most commonly used grip. In this figure, (b) shows a very unusual writing grip. In this case, it would be challenging to render an accurate model of the user hand.

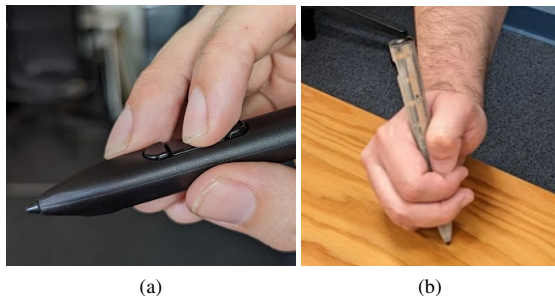


Figure 5: (a) Common in-air stylus grip (b) Unusual 2D writing grip

As a result, many apps show the stylus model only, without showing the hand. The Vermillion app is an interesting exception [17]. It has a feature within the app that allows the user to dynamically

adjust the position of the virtual hand to be closer to their real world physical grip.

3.4 Alternatives to Controller Chords

Chords allow the use of two inputs on the same controller at the same time. There is a learning curve associated with these techniques and they also require a certain level of manual dexterity. However, they can also be a very powerful feature for advanced users. Several of the more complex design and creativity apps support such chords, e.g., press-and-hold the Grab button and then press the Trigger to duplicate an object in ShapesXR. Because these chords are often considered advanced features, they can also be performed via the app UI, although this will be slower.

This powerful multi-digit chord capability is not easily possible on the MX Ink stylus. As a result, this means utilizing the UI method (as opposed to the quick chord option). Developers also remap some of that functionality to unused non-dominant hand Touch controls when possible.

3.5 2D Surface Alignment

One of the advantages of a stylus form factor is the ability to write, sketch and interact with physical objects; primarily (but not always) on a 2D planar surface. The Quest Pro Controller also has a pressure sensor in the base of the handle grip, but it requires frequent adjustment of the user grip to use this feature. For these 2D planar surface interactions, the accuracy and persistence of the alignment of the virtual and physical surfaces is critical to a good experience. Several different approaches have been proposed and evaluated by Kern et al. [11], and they have made the source code publicly available.

Together with Meta Quest platform spatial anchors, and libraries like Photon Engine [19], this can enable the development of robust, multi-user mixed reality virtual whiteboards with good quality sketching and writing capabilities on physical 2D surfaces. An example is shown in Fig. 6; a virtual blue color note is created in-air, and another yellow color note is already pinned to the wall surface.



Figure 6: Notes created in air, and posted on wall

The above Fig. 6 shows a virtual Post-it note being created in air. However, there are also demonstrated benefits of writing on a 2D surface, as shown by Drey et al. in their VRSketchIn project [8], and Arora et al. with SymbiosisSketch [2]. Such notes could be created in-air, or on a 2D desk or wall surface.

3.6 Absence of Stylus and Inking SDKs

Over time, platforms that support a stylus have created SDKs and support libraries that simplify the task of adding support in apps. For example, Microsoft Windows Ink provides support for creating digital handwritten notes, drawings, and annotations. Apple PencilKit makes it easier to incorporate hand-drawn content into apps on iPadOS or MacOS. The exact features vary by platform, but typically provide a framework for processing the input stream, managing and rendering ink data as strokes on the output device.

Without these frameworks, it is a significant effort for an individual app to develop user experiences that resemble writing or drawing on paper with a pen, pencil, or brush. Beyond planar 2D surfaces, there are additional complications associated with inking in 3D space. Currently on XR platforms, these types of capabilities are implemented on a per-app basis rather than as part of platform tools. This is currently a significant barrier to app developers, i.e., it would be nice to have good quality ink strokes, brushes, etc. However, it is a significant development effort to create these from scratch.

3.7 Interaction Toolkits

There are several XR interaction frameworks that can greatly simplify app development. For example, the Meta XR Interaction SDK for Unity supports object grab and scale, push buttons, teleportation, UI panel navigation, and more while using Touch controllers or tracked hands. Other examples are Unity's XR Interaction Toolkit and the Microsoft-driven open source project MRTK3. These toolkits were developed with a focus on controllers and hands, and require some updates to better support stylus interactions, e.g., adjustments to better cater for the different OpenXR grip and aim pose definitions for a stylus [12].

4 CONCLUSION

Most mainstream commercial HMDs have shipped with their own platform-specific 6DoF controllers. In the past, there were notable differences in these different 6DoF controllers, but in recent years the Meta Quest Touch Controller feature set has emerged as a de facto standard. There are notable benefits for users, familiarity with one controller means it is easier to use a different (but similar) controller on another platform. There are also advantages for developers as it (somewhat) standardizes the interaction mappings across different vendor platforms.

The Logitech MX Ink product was released in September 2024. It extends the Meta Quest platform with the addition of a 6DoF stylus. This new input device opens up new opportunities, but also introduces challenges to existing paradigms, which have been primarily focused on the default Touch Controllers and tracked hand interactions.

The MX Ink stylus is natively supported in the Meta Quest platform; no additional software components need to be installed. However, as described in Section 2, there are still a few additional steps required for a developer to add basic stylus support to their app. To fully embrace the benefits of a stylus, there are additional design and interaction considerations, as outlined in Section 3. This includes supporting handedness, considering the visual representation of the stylus and hand in the app, and dealing with an input device with a control set different from the default Touch controller. In the future, to better support developers, it is required to enhance existing interaction frameworks with stylus support and have robust shareable samples for 2D surface alignment and inking.

The release of the original HTC Vive platform prompted the development of several commercial MR tools that initially focused on ideation and sketching. Recent years have seen the expansion of the commercial MR tools market, with the addition of more capabilities such as 3D modeling, support for lighting, texturing, and high-quality rendering for review purposes. These tools are gradually filling in the gaps required to enable end-to-end 3D content

creation workflows in MR environments [9]. Logitech MX Ink is also a valuable addition to this ecosystem, enabling more research and development in the MR design and creativity space.

Beyond the design space, MX Ink support has been added in applications for a range of additional use cases, including collaboration, medical, education, training, and games.

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